

Study Questions, by Topic (cumulative), will be added here as the term progresses

I. Introduction

1. For each of the following statements, explain whether they are normative or positive

a) Government expenditure on road construction should decrease unemployment rates.

Normative/Positive	Explanation

b) Government expenditure on road construction is predicted to increase unemployment rates.

Normative/Positive	Explanation

c) The government of [Vulcan](#) spends too much on road construction.

Normative/Positive	Explanation

d) The temperature today is too warm for it to snow.

Normative/Positive	Explanation

e) There is too little snow fall (.2 cm) to justify salting the roads.

Normative/Positive	Explanation

f) Canadian GDP is currently growing at 8% per year.

Normative/Positive	Explanation

g) Monetary policy needs to focus on inflation to be effective.

Normative/Positive	Explanation

2. For each of the normative statements in question 1, transform those statements into positive ones (you may/not need as many rows as I have provided) **Many transformations are acceptable.**

Original Statement (normative):	Transformed Statement (positive):

3. For each of the positive statements in question 1, transform those statements into normative ones (you may/not need as many rows as I have provided) **Many transformations are acceptable.**

Original Statement (positive):	Transformed Statement (normative):

4. Suppose you publish a research report indicating that more generous (larger) [Employment Insurance](#) payments will reduce workers' incentive to find new jobs quickly, which will result in higher unemployment rates.

a) Is this reported outcome written in a normative or positive way? Explain.

Normative/Positive (underline one)	Explanation:

b) Is there any reason why you might want to use some normative statements when describing the implications of your research?

Yes/No (underline one)	Explanation:

c) What additional statement(s) might you wish to make in this context? **Many good answers possible**

Additional Statement(s):

5. Suppose that you are working as a policy advisor for the Government of [Rohan](#), and you design a model to predict the implications of minimum wage increases on employment levels. Suppose that the model you design is a market clearing model that assumes all workers are identical and all firms are identical. Your model predicts that when minimum wage is set above the market clearing wage, employment levels (N) falls. Your model also predicts that when minimum wage is above the market clearing wage that unemployment will rise, but when minimum wage is at or below the market clearing wage, that unemployment will be zero.

a) Using the Model Evaluation criteria from our lecture slides, explain whether your model meets each criterion: "Well", "Somewhat", or "Not at All"

Criterion	Meets criterion “Well,” “Somewhat,” “Not at All”	Explanation
Realistic Assumptions		
Understandable & Manageable		
Testable Implications with Available Data		
Predictions Consistent with Data		

b) Based on the evaluation of your model against these criterion, is your model a reasonable one to advise policy makers in the Government of Rohan on the implications of minimum wage on employment levels? Explain why or why not.

Reasonable? Yes/No/Uncertain (underline one)	Explanation:
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c) Would you recommend considering other models for this policy question? Why or why not?

Other models? Yes/No (underline one)	Explanation:
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6. Explain one benefit of using algebraic expressions or mathematical characterizations of behaviour/economic structure/constraints as we tend to do in theoretical analysis in Economics.

Benefit:	
Explanation:	

II. Measurement, Accounting and Data

7. Suppose you are hired by the government to assess the impact of the latest (COVID) recession on household well-being using detailed GDP measures from Statistics Canada over a 5 year period.

a) Which method(s) of GDP calculation would you use, and why?

Method(s):	
Explanation:	

b) Would you use Real or Nominal measures of GDP, and why?

Real or Nominal:	
Explanation:	

c) Which inflation index would you use and what are the drawbacks of this measure for your task?

Index:	
Explanation:	

8. Explain whether the items below will be included in Canada's GDP, GNP, both, or uncertain:

a. Tuition expenditures of Irish students in Canada

Answer:
Explanation:

b. Canadian singer, Anne Murray's, concert in Vienna

Answer:
Explanation:

c. Louise Penny's sale of her latest Three Pines Quebec Mystery Novel

Answer:
Explanation:

9. One of your class mates claims that Gross Domestic Product is an unreliable measure of aggregate economic activity. She says that GDP will underestimate actual economic activity because much intangible production is not included. She further claims that exclusion of the underground economy (drug sales, prostitution income) also aggravates the underestimation.

a. Is it possible that GDP could underestimate economic activity? Explain

Answer: Yes/No	
Explanation	

b. Explain why GDP may actually be over-estimated, rather than underestimated.

Explanation	
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10. For each of the following scenarios, will our measure of GDP over-estimate or under-estimate aggregate economic activity. Explain.

a) Cleaning your apartment/dorm room. (2)

Answer:

Explanation:

b) Cleaning up municipal water that had been contaminated from agricultural run-off. (2)

Answer:

Explanation:

c) Taking your cousin to pet the Llamas in Waterloo Park. (2)

Answer:

Explanation:

11. The [Mrs Ws](#) sell teleportation devices which they purchase from a manufacturer for \$100,000. The Mrs Ws pay their employees a total of \$40,000, pay \$25,000 in taxes and have profits of \$5,000. What is the value added to the economy by the Mrs Ws? Show your work.

Value Added:

Calculation:

12. Answer the following questions and provide explanations for each answer:

a. Statistics Canada adopted chain weighting in May 2001. This change led to a downward revision in investment rates (the rate of change of the capital stock), because the fall in computer prices gives the growth in the volume of computing equipment a smaller weight when more recent prices are used to construct real investment I. What effect will this revision have on measured TFP growth in Canada?

Effect on measured TFP Growth:	
Explanation:	

b. [Anyanwu's](#) medicinal flower shop sells \$40 worth of flowers (including taxes), to its customers, and pays \$10 to the farmer for the flowers, \$5 to the government in taxes, and \$20 to a retail clerk. The farmer, in turn, pays a farm hand \$2 to cut the flowers and pays \$1 in taxes. The retail clerk pays \$6 in income tax, while the farm hand pays \$0 in income tax. Use the income approach to calculate GDP from Anyanwu's. Show your calculations.

GDP using Income Approach:	
Calculations:	

c. If Government debt started at zero and for 10 years government expenditure exceeds revenue by \$4, calculate the government deficit in year 10 and the government debt at the end of year 5.

Deficit in year 10:	
Debt in year 5:	

13. Given the following information about Spooky Island that sells jars of Protoplasm

	Year 1	Year 2	Year 3
Quantity of Protoplasm Jars Sold	400	435	500
Price of Protoplasm Jars	1.00	1.15	1.20

a. Calculate Nominal GDP and the GDP deflator for years 2 and 3 assuming year 1 is the base year. Explain and show your work.

	Year 2	Year 3
Nominal GDP		
GDP Deflator		
Calculations:		

b. Calculate the inflation from year 2 to 3. Show your work.

Inflation year 2 to 3:	
Calculation:	

c. Assume jars of protoplasm do not rot from one year to the next. Also assume that you don't care whether you consume your jars of protoplasm in year 2 or 3. If the nominal interest rate in year 2 is 4%, would you be better off loaning \$115 in year 2, borrowing \$115 in year 2, or be equally well off regardless? Why?

Better off loaning/borrowing or equally well off:	
Explanation:	

d. Now suppose that nominal interest rates fall to 3%, but jars of protoplasm go bad if you don't consume them right away, and you don't have capacity to consume more than 100 jars a year. If you had \$230 in year 2, and spent 115 of it on jars in year 2. Would you be better off loaning your remaining \$115 in year 2, or saving the cash in your pocket? Explain.

Better off lending or keeping in pocket:	
Explanation:	

III. Economic Frameworks

14. Suppose that you wish to design a model that would allow policy makers to predict the impact on unemployment that would arise from an increase in the compensation for lost earnings (EI payments) under the [Employment Insurance](#) program.

a) Explain the type of model you would like to use, according to the following classifications: static versus dynamic, general versus partial equilibrium, and a market clearing versus a non-market clearing model. Be sure to defend your choices.

Choice (<u>underline one</u>)	Explanation:
Static vs Dynamic	
General vs Partial	
Market Clearing vs non-clearing	

b) Would unemployment be an exogenous or endogenous variable in your model? Explain.

Unemployment Endogenous or Exogenous:	
Explanation:	

15. Suppose an economy has the production function $Y = AK^{0.25}N^{0.75}$

In 2021: Real GDP = 74,802; K = 268,147; N = 2414

In 2022: Real GDP = 78,561; K = 301,262; N = 2632

a. Calculate total factor productivity (TFP) for 2021 and 2022.

TFP 2021:	TFP 2022:
Calculations:	

b. What is the growth of total factor productivity?

Growth of TFP:	
Calculation:	

c. Calculate the percent increase in real GDP from 2021 to 2022.

Percentage increase in real GDP:	
Calculation:	

d. Suppose productivity remained constant from 2022 to 2023, but K and N both triple during that time. What will GDP be in 2023?

GDP:	
Calculation:	

16. Suppose the country of [Narnia](#) has the following output and input levels for 2023 and 2024:

Year	Y	N	K
2023	20	1	4
2024	1000	4	25

a) If Economists believe that the production function for Narnia's economy is $Y=AKN^{0.5}$, what is TFP growth for 2023-2024?

TFP Growth:
Calculation:

b) Now suppose that the true production process in Narnia is characterized by $Y=AK^{0.5}N^{0.5}$. Calculate actual TFP growth.

TFP Growth:
Calculation:

c) What do our results from parts a) and b) imply about our accuracy in measuring TFP growth?

Explanation:

17. Suppose that city of [Willowbee](#) has a production function $Y = AK^{0.5}N^{0.5}$, where $A=100$ & $K=1$.

a. Graph this production function with respect to changes in Labour input (show at least 2 accurate points & label all axis & curves). (3)

Production



b. What does the MPN represent intuitively and graphically? (2)

Answer:

c. What is the marginal productivity of labour at $N=9$, assuming A and K are fixed at 100 and 1 respectively. (Note: if it helps, you can say $8^{0.5}=2.8$). (2)

Answer:

Calculations:

d. How would MPN be affected by an adverse supply shock? Explain. (2)

Answer:

Explanation:

IV. Labour Market, Employment, and Unemployment

18. Briefly answer the following questions.

a. Explain what is meant by diminishing marginal productivity of a factor?

Explanation:

b. What factors (list at least two) cause labour demand to increase?

Factor1:

Factor2:

c. What factors (list at least two) cause labour supply to decrease?

Factor1:

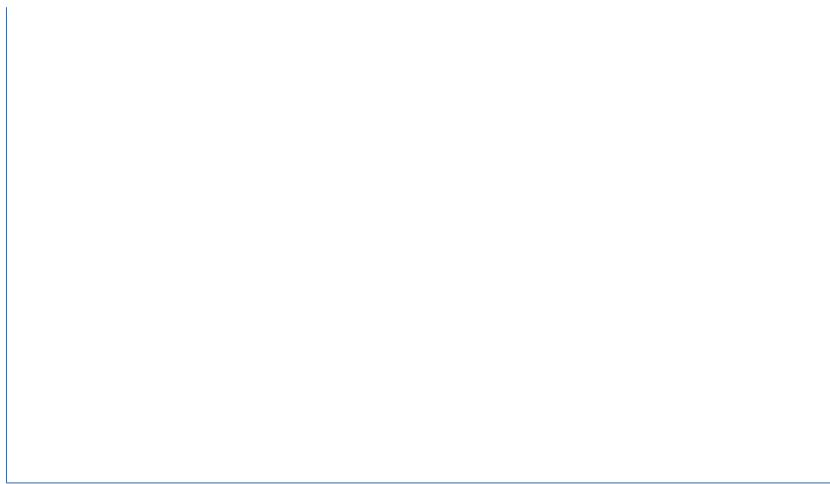
Factor2:

19. Suppose that the country of [Florin](#) has 100 workers, each with a labour supply function of $N^s=3w$.

a) Construct and graph the aggregate labour supply curve for Florin. Be sure to label both axis.

Labour Supply curve:

Labour Market



b. Now suppose the market wage, $w^* = 20$, how much labour will be supplied in Florin?

Answer:

Calculations:

c. If the government imposes an income tax, $t=.5$, such that workers only retain 50% of their paid wages, explain and show graphically how the labour supply curve is affected (don't worry about what happens to w^*). Derive the new slope of the labour supply curve.

Answer:

New slope:

Explanation:

20. Suppose firms' hourly marginal product of labour is given by $MPN = A(350 - N)$, where $A=.3$

a. How are real wages determined in the labour market? Explain how the MPN function is related to the labour demand function and graph the labour demand curve using at least 3 points.

How Real Wages are Determined:

How MPN is related to labour demand:

Labour Market



b. Let the real wage be fixed at \$15/hour. How much labour will the firm want to hire? Show your work.

Labour Demand at \$15/hr:

Calculation:

c. Suppose real wages are no longer fixed, but are determined in the labour market equilibrium. Further suppose the aggregate supply of labour is $N^s = 200 + 22w$. Find the equilibrium levels of employment (N) and real wage rate w . Graph the labour market.

W^* :

N^* :

Calculation:

d. What would be the impact on the economy of the government fixing the real wage at \$15? Explain & illustrate this on your graph.

Impact:	
Explanation:	

21. Suppose our economy consists of one firm, WeCreateJunk, with $MPN=100-\frac{1}{2}N$.

a) If there are 2 workers in our economy, each with a labour supply function of $N_i^S=w$, where $i=1,2$ represents each worker. Solve for the equilibrium labour market wage and employment levels. Show your work!

Equilibrium wage, w^*:	
Equilibrium employment, N^*:	
Calculations:	

b) Graph, below, the labour demand and supply functions and label the equilibrium point E_1 . (Be sure to label both axis)

Labour Market



c) If an income tax of $\tau\%$ were imposed on wages (assume τ is between zero and one), would the equilibrium wage and employment be higher or lower? Explain. Show this change using the labour market graph above. Label, but do not calculate the new equilibrium. (2)

New w^* higher or lower?	
New N^* higher or lower?	
Explanation:	
Don't forget to show Show new w^* and new N^* on graph & label new equilibrium	

d) How would you re-write the new labour supply function to incorporate the tax?

New N^S:	
Calculations:	

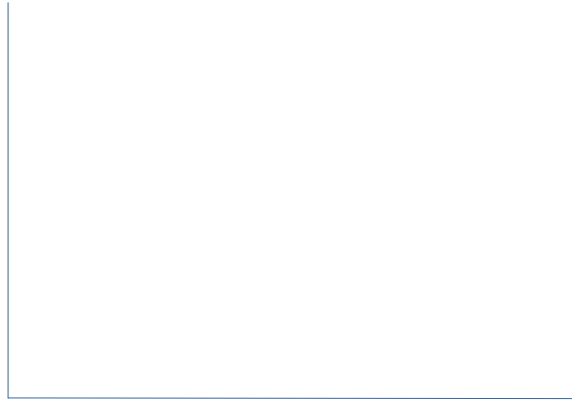
22. Suppose that the country of [Mulgore](#) produces hay and has marginal product of labour function $MPN = A \cdot 540 - 2N$

a. If $A=2$, what will be the demand for labour in Mulgore when real wage =20?

N^D:	
Calculations:	

b. Given $A=2$, graph the demand curve for labour (show at least 2 accurate points).

Labour Market



c. There are three workers (hay bailers) in the economy.

One has a supply function $N = \frac{1}{2} w$,

another has supply function $N = 4 + \frac{1}{2} w$,

and another has supply function $N = -4 + w$

Calculate the aggregate labour supply (write down the actual function) and graph it, showing at least 2 accurate points.

N^s:	
Calculations:	

d. Calculate the labour market equilibrium quantity of labour and wage and graph this point (call it E).

Be sure to label the equilibrium quantity of labour and wage with their exact values.

N*:	
w*:	
Calculations:	

23. The following macroeconomic information is available for the Country of [Rivia](#)

	2022	2023
Output (books)	9000	8000
Employed	900	800
Unemployed	100	100
Price per Book	9.00	8.00

Calculate the following and be sure to show your work, including the formulas used in the calculation.

a. Growth rate of nominal GDP from 2022 to 2023

N^s:	
Calculations:	

b. Growth rate of books produced from 2022 to 2023

N^s:	
Calculations:	

c. Growth rate of real output from 2022 to 2023 in 2022 dollars

Growth in real 2022 dollars:	
Calculations:	

d. Growth rate of real output from 2022 to 2023 in 2023 dollars

Growth in real 2023 dollars:	
Calculations:	

e. Why is growth rate the same regardless of which base year you use?

Answer:

f. Calculate the unemployment rates for 2022 & 2023, and show your work.

Unemployment rate 2022	
Unemployment rate 2023	
Calculations:	

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24. For each of the following, explain whether the measured amount represents a stock or a flow, and explain:

a) Saving increases by \$180 billion in 2020.

Stock or Flow:
Explanation:

b) Canadian merchandise exports fell by 12% in 2020.

Stock or Flow:
Explanation:

c) Unemployment rose to 1 million people in Ontario by the start of May, 2020.

Stock or Flow:
Explanation:

d) The value of total household wealth in the U.S. had dropped to about 54 trillion by the start of the third quarter of 2008.

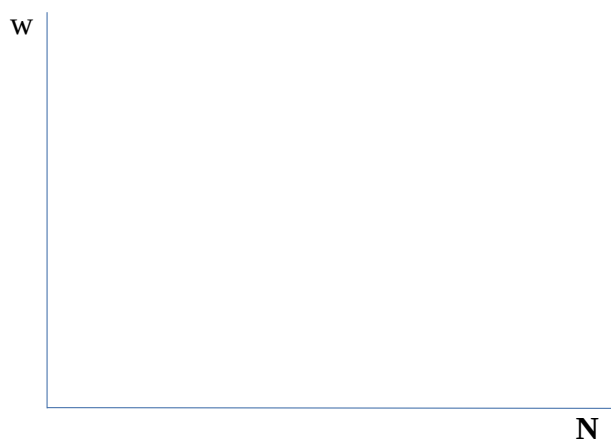
Stock or Flow:
Explanation:

25. Suppose that all firms are identical and all workers are identical, and that aggregate labour demand is $N^D = 10 - \frac{1}{2} w$. Further suppose the aggregate supply of labour is $N^S = 2 + \frac{1}{2} w$.

a) Find the equilibrium levels of employment (N) and real wage rate w . Show your work. Graph the labour market.

N*:	
w*:	
Calculations:	

Labour Market (not to scale)



b. Would the level of employment and level of unemployment be higher or lower in this economy if the government fixed the minimum wage at \$10? Illustrate this on your graph from part a). Assume that $P=1$, so that real and nominal wage are the same.

Employment:	
Unemployment:	
Explanation:	
Don't forget to show on graph	

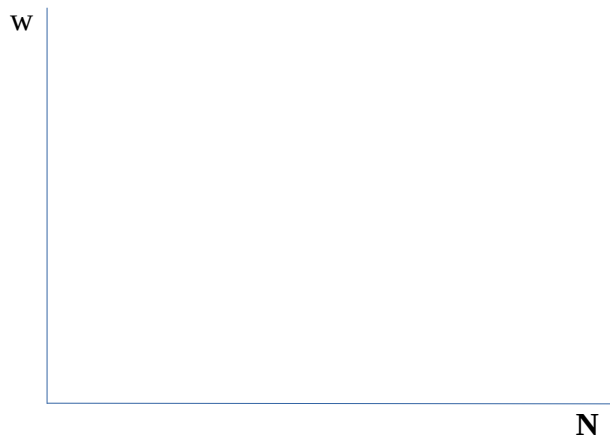
c. Suppose that minimum wage remains fixed at \$10, and price level is unchanged at $P=1$, but that the economy grows, with more identical firms entering the market. Will this model predict unemployment rises, falls or stays the same? Explain and show graphically using your graph from part a).

Unemployment:	
Explanation:	
Don't forget to show on graph	

26. Suppose we are in an economy where real minimum wage (nominal minimum wage divided by P) is above w^*

a) Graph the labour market and show the level of unemployment in your graph.

Labour Market (not to scale)



b) Indexing minimum wage to inflation is becoming more common. A province that indexes minimum wage to inflation will increase the minimum wage at the same rate as inflation so that the real minimum wage does not fall. Suppose that the price level rises dramatically from one year to the next. What does the model predict will happen to the unemployment level for a province that does not index their minimum wage (case I), versus a province that does index their minimum wage (case II), relative to part a)?

Unemployment (case I):	
Unemployment (case II)	
Explanation:	
Don't forget to show on graph	

27. Answer the following questions:

a) Define Full-Employment Output

Answer:

b) Explain what is Frictional Unemployment

Answer:

c) Explain what is Cyclical Unemployment

Answer:

d) Explain what is Structural Unemployment

Answer:

e) Which type(s) of unemployment (among the three you discussed above) were most likely to be higher during the height of regional lockdowns in 2020? Explain.

Answer:

Explanation:

f) Which type(s) of unemployment (among the three you defined above) were most likely to be higher shortly after regional lockdowns were relaxed in 2021 and the economy was rebounding? Explain.

Answer:

Explanation:

g) Would you expect the official unemployment rate to be a good measure to track changes labour market outcomes as firms shifted to posting more part time and contract jobs? Explain.

Answer:

Explanation:

28. Suppose that you are using a Search and Matching Framework to understand unemployment levels in the economy. Assume that you have designed a general matching function $M(U,V)$ which predicts the number of matches formed increases with the number of workers searching and the number of job vacancies posted, there is free entry into and out of the market, and you assume a fixed job destruction rate of δ (which is a fraction between zero and 1). Also suppose that you assume a fixed labour force (LF) where the level of unemployment is then defined as $U=LF-E$, where E is the number employed.

a) What condition do you use to calculate the steady state level of employment and unemployment in the economy? Explain.

Answer:
Explanation:

b) If there is an increase in the labour force (due to an increased participation rate, say), will your model predict the steady state level of employment to rise, fall, or stay the same? Explain.

Answer:
Explanation:

c) If match productivity (output from a match) rises, and the labour force remains constant, will your model predict a rise/fall or no change in the steady state level of unemployment? Explain.

Answer:
Explanation:

d) If the job destruction rate in the economy were to rise, will your model predict a rise/fall or no change in the steady state level of employment? Explain.

Answer:

Explanation:

e) For parts b), c) and d) will the change happen instantly or gradually over time? Explain (be sure to refer to changes in flows).

Answer:

Explanation:

f) If you had a set of algebraic equations to fully specify this model, would you feel more comfortable or less comfortable talking about the model's predictions? Explain.

Answer:

Explanation:

V. Consumption, Saving and Investment (for this entire section, assume a closed economy)

29. Answer the following questions:

a) Explain why Prudence may wish to smooth her consumption over time, and how Prudence might go about doing so.

Explanation of Why & of How:

b) Explain the why the substitution effect might cause individuals to decrease current period consumption when the real interest rate rises. Then explain the income effect for borrowers when said interest rate rises. Do these work in the same or opposite directions?

Substitution Effect Explanation:

Income Effect for Borrowers Explanation:

Same or Opposite direction:

c) Of the following cases, explain which are consumption smoothing behaviours and which are not.

i) Real interest rates rise and Prudence (who has interest deferred student loans decreases her current consumption slightly and expects to reduce her future consumption slightly as well.

ii) Prudence receives a bonus of \$50 today and decides to spend it on a \$50 bottle of Canadian Whisky to share with her friends tonight.

iii) Prudence expects a raise of \$10,000 next year, so she decreases her savings slightly in the current period to purchase slightly more groceries.

Answer:

Explanation:

30. Explain and graph the impact on the saving demand curve and the equilibrium r^* of the following changes:

a) Current Y falls because everyone has caught this flu and is less productive!



Impact on S^D:
Impact on r^*:
Explanation:

b) Expected future Y falls.



Impact on S^D:
Impact on r^*:
Explanation:

c) A new government is elected that wants to balance the budget by cutting spending.



Impact on S^D:
Impact on r^*:
Explanation:

31. Explain and graph the impact on the investment demand curve, equilibrium r , S and I , of the following changes:

a) A technological breakthrough increases MPK^f



Impact on I^D:
Impact on r^*:
Impact on S^* & I^*:
Explanation:

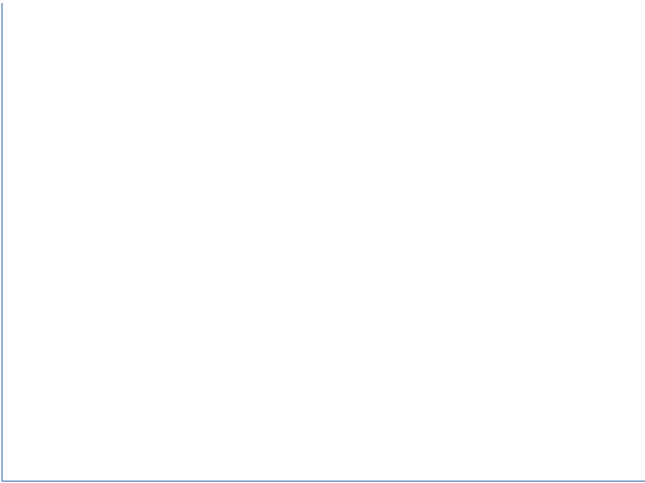
b) Effective tax rate on capital, τ , decreases.



Impact on I^D:
Impact on r^*:
Impact on S^* & I^*:
Explanation:

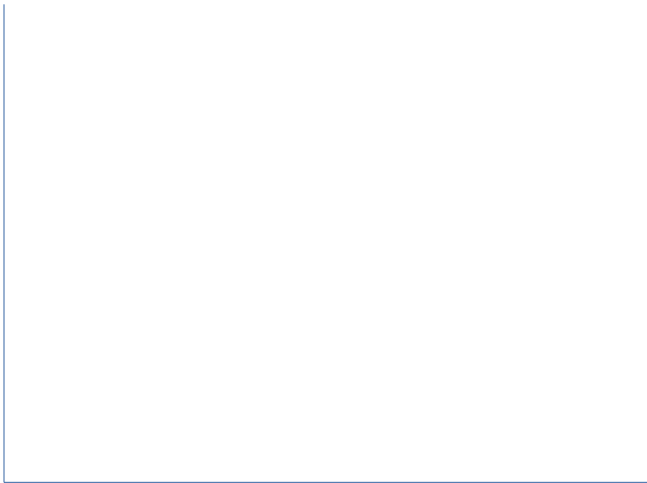
32. Explain and graph the impact on the goods market (curves and equilibria r^* , S^* , I^*) of the following changes:

a) Increase in Y^f .



Impact on I^D:
Impact on S^D:
Impact on r^*:
Impact on S^* & I^*:
Explanation:

b) Temporary rise in MPK and Pk simultaneously.



Impact on I^D:
Impact on S^D:
Impact on r^*:
Impact on S^* & I^*:
Explanation:

33. Consider a closed economy and the following output, desired consumption and desired investment:

$$Y = 1000 \text{ (full employment output)}$$

$$C^D = 300 + 0.3Y - 175r$$

$$I^D = 500 - 250r$$

$$G=0$$

a) Find the real interest rate that clears the goods market and show your equation for desired savings.

Equation for S^D:
r^*:
Explanation:

b) Suppose now, we introduce $G = 100$. What is the effect on r^* and how is the desired savings equation changed?

Equation for S^D:
r^*:
Explanation:

VII. Money and Inflation

34. Suppose that the country of Minionland is deciding whether to use Bananas and Beads as their currency. Assuming that either would be an acceptable form of payment for goods and services, explain which of Bananas or Beads would be the more suitable choice for their currency.

Bananas or Beads more suitable currency:
Explanation:

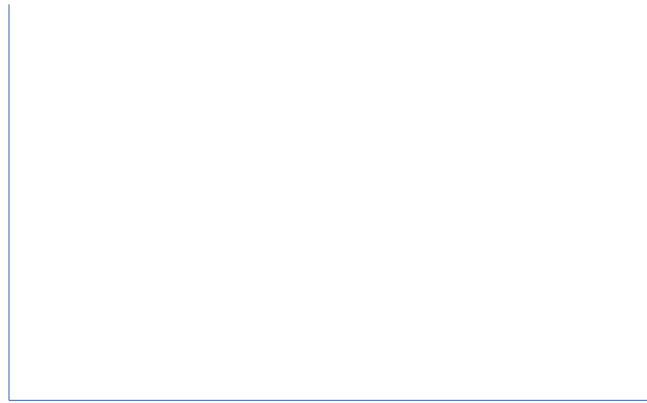
35. Show graphically and explain the impact of the following on the money demand and/or money supply curves, and the equilibrium real interest rate in the Asset Market.

a) Increase in liquidity of stocks & bonds, impact on $L(.)$ and r^*



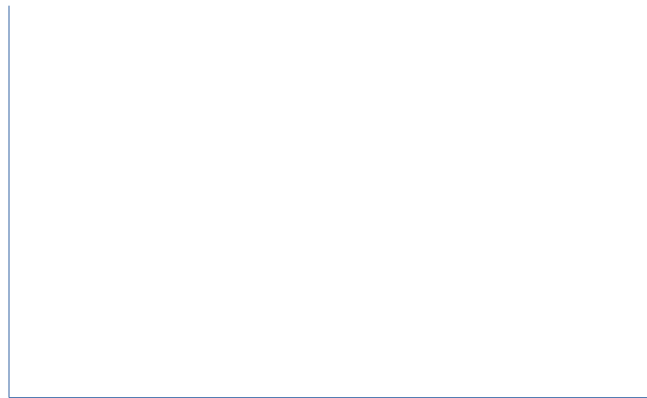
Impact on $L(.)$ and or M/P:
Impact on r^*:
Explanation:

b) Contractionary monetary policy (to try to curb inflation) causes M_s to fall --show graphically, & what is impact on r^*



Impact on $L(.)$ and or M/P:
Impact on r^*:
Explanation:

c) People begin to distrust banks and the Currency/Deposit ratio of public rises.



Impact on $L(.)$ and or M/P:
Impact on r^*:
Explanation:

36. Answer the following questions

a) Economists often refer to tradeoffs in decision-making, which tradeoff do Okun's Law and the Phillip's curve model?

Answer:

Explanation: .

b) Explain why the Bank of Canada uses and continues to use a 2% inflation target?

Explanation:

c) Describe three real-world examples of the costs of inflation

Answer:

d) Using what you have learned about the costs of unemployment, explain how unemployment has both individual and aggregate impacts.

Answer:

e) Are the costs of inflation and unemployment temporary or permanent or both? Explain.

Answer:

37. Consider how our fractional reserve banking system influences money supply via the multiple expansions of loans and deposits.

a) If the public decides to keep more cash in pocket, will this decision increase or reduce the effect of the fractional reserve system on the expansion of money supply? Explain intuitively and show algebraically.

Answer:

Explanation:

b) What are the implications of a higher reserve deposit ratio on the money supply? Explain.

Answer:

Explanation:

c) From a bank's perspective, list one advantage and one disadvantage of a low reserve deposit ratio.

Advantage:

d) What would happen to money supply if the reserve deposit ratio were increased to 1 (100%)?

Answer:

Explanation:

38. What is the asset market equilibrium condition?

Answer:

39. Suppose that in the country of Vanyassa the public decides to keep all currency in pocket and not use banks. What is the money supply in this economy?

Answer:

Explanation:

40. Answer the following questions:

a) Define the money multiplier

Money Multiplier:

b) What is the value of the money multiplier in a system of 100% reserve banking, that is $res=1$?

Answer:

Explanation:

c) Given: $M=1,000,000$, $CU=400,000$, and $res=0.3$, what is the value of the money multiplier, bank reserves, BASE, and deposits?

Answer: Deposits =

Explanation:

Answer: Reserves =

Explanation:

Answer: Money Multiplier =

Explanation:

Answer: BASE =
Explanation:

d) Suppose there are three countries (A, B, and C). Each country has a BASE of 1 million. Which country will have the greatest money supply – Country A, which has no banks, and only currency, Country B which has banks, where people hold deposits, but the reserve deposit ratio is 100%, or Country C where there are banks and people hold deposits, and the reserve deposit ratio is 50%?

Answer:
Explanation:

41. Describe the effect of each of the following on M:

a) An increase in muggings causes people to deposit more of their cash in their bank accounts.

Answer:
Explanation:

b) Increased stability causes banks reduce their reserve deposit ratio.

Answer:
Explanation:

c) The government buys government bonds from the Bank of Canada with newly printed currency

Answer:
Explanation:

VI. Trends, Cycles and Inequality

43. Answer each of the following questions:

a) Explain (in detail) why it is difficult to determine whether real wage is cyclical or not.

Explanation:

b) Are durable consumption good more or less pro-cyclical than non-durable consumption goods? Explain.

Answer:

Explanation:

44. Answer the following questions on business cycles:

a) How are business cycles measured?

Answer:

b) Why is it difficult to define recessions

Answer:

c) list 3 ways you might quantify a recession.

Answer:

d) If you had to choose between RBC and sticky wage approaches, which would be better equipped to try to understand the labour market impact of business cycle fluctuations? Explain.

Answer:

Explanation:

45. Suppose that the Bank of Canada implements policy that increases money supply. Assume some price movement but complete wage rigidity in this time frame (short run), and assume the initial $W/P > w^*$, to answer the following questions.

a) Show the impact of this policy in the Asset Market, Goods Market and Labour Market.

Goods Market	Asset Market
Description:	

b) In the short run, when wages are sticky, explain what happens to Employment, Saving, real interest rates, GDP and prices.

N:
S:
r:
Y:
P:
Explanation:

46. Answer the following questions related to Business Cycle Transmission (uses some open economy concepts):

a) Your friend is concerned about a recession in the UK having implications on the Canadian Economy. Explain the likely impact on the Canadian economy of a recession in the UK.

Explanation:

b) Your friend has now started worrying about the potential decline in growth of China's economy. Explain the likely impact on the Canadian economy of economic contraction China.

Explanation:

47. Your friend pays attention to many economic indicators, and has predicted an imminent recession due to a decline in the average work week (hours) in Manufacturing.

a) What concerns might you raise about relying on this variable to predict recessions? You may wish to use FRED data to support your statements.

Concerns:

b) List two reasons why it is difficult to predict recessions, in general.

Reasons:

42. Graph and explain the impact of a positive supply shock (rise in A) on N^* , w^* , Y^* , r^* , P & I^* , assuming a market clearing long run framework (no sticky wages/prices).

Labour Market	
Goods Market	Asset Market
Explanation:	
N*:	
w*:	
Y*:	
r*:	
I*:	
P:	

43. Graph and explain the impact of decrease in G on S^* , I^* , r^* , Y , P , N & w in sticky wage framework.

Goods Market

Asset Market

Explanation:

N:

w:

Y:

r^* :

I^* :

P:

48. Answer the following questions on economic growth:

a) List three major input variables associated with growth.

3 input variables associated with growth:

b) Suppose that the Country of Lomond is identical to the Country of Renfrewshire in all respects except that Lomond has lower GDP due to its lower levels of Capital stock. So both countries have the same rates of saving and of depreciation, but different levels of capital and output. Assuming that neither Lomond nor Renfrewshire have yet reached their steady state balanced growth path (they remain below it), which country would you expect to grow faster over the next year?

Faster Growth:

Explanation:

c) Discuss two negative aspects of growth, when growth is driven by technological improvement.

Negative aspects of growth:

d) Continuing from part c, explain one policy that a government might implement to mitigate the negative aspects of technology driven growth.

Policies to mitigate negative aspects:

49. With the increasing capabilities and use of AI (Artificial Intelligence), some individuals have raised concerns over job loss.

a) Why might skilled labour have good cause to worry about job loss with increases in AI?

Cause for Concern:

b) Explain one potential benefit to AI for skilled and unskilled workers. Why might skilled labour benefit from AI?

Benefits of AI:

50. Answer the following questions on measuring poverty:

a) Why is it difficult to define poverty? Explain two reasons.

Difficulties defining poverty:

b) Explain why poverty is a subjective and a relative measure.

Poverty is Relative because:

Poverty is Subjective because:

51. In 2018, the Canadian Government initiated a poverty reduction strategy based on the Market Based Measure (MBM) of poverty. Suppose the poverty rate is measured as the percent of families with family income below the poverty line as measured by the MBM.

a) Suppose that the number of families in Canada remains constant between 2025 and 2026, and that the poverty line remains constant between 2025 and 2026. Further suppose that 100,000 families with income below the MBM in 2025 experience a further decline in income in 2026, but no other families in Canada experience a decline in income, would the poverty rate change in Canada? Explain why or why not. (Do in class some examples like this – both decrease and increase in income).

Will the poverty rate change?:

Explanation:

b) Suppose poverty rates fall by 10% in 2027, and the government claims that this decline indicates the effectiveness of the poverty reduction strategy. Explain what concern(s) you might raise about this claim.

Concerns:

c) In addition to income based measures of poverty and inequality, economists consider additional measures and dimensions. List three alternative measures/dimensions which are important to consider for a fuller picture of poverty and inequality. Describe why each are important.

3 alternative measures/dimensions:

Explanation:

52. List two benefits and two drawbacks of using generalized entropy measures (e.g. Theil index) rather than the Gini to measure inequality.

3 alternative measures/dimensions:

Benefits:

Drawbacks:

VIII. Monetary and Fiscal Policy

53. Answer the following questions on monetary and fiscal policy:

a) Describe two reasons economists might support the rules approach to monetary policy

Drawbacks:

b) Discuss three limitations to conducting monetary policy

Drawbacks:

c) Describe two drawbacks of fiscal policy:

Drawbacks:

54. Suppose the new government of Spartinth is considering three policies which would increase Government Spending: 1. building new roads to facilitate traffic, 2. a food relief program -allocating funds to people whose income falls below a specific threshold, 3. Provision of a lump sum increase in funding to public universities for the current year.

a) Which policy/policies represent an automatic stabilizer and why.

Automatic Stabilizer(s):

Explanation:

b) Do any of these policies have potential negative effects on economic growth or fluctuation? Explain.

Answer:

IX. International Economic Concepts, Exchange Rates and Balance of Payments

55. Individuals sometimes raise concerns about the trade deficit being “too large” or “too persistent.”

a) What is the trade balance? And how does it differ from the Current Account balance?

Trade Balance:

Difference between Trade Balance and CA balance:

b) If a country has a negative CA balance, is it a net borrower or a net lender?

Answer:

Explanation:

c) Explain why a negative CA balance can be a good thing.

Benefits:

d) Describe one concern with having a persistent negative CA balance?

Concern:

56. Many countries with low GDP have borrowed heavily. This borrowing, as discussed in question 57c), can be a good/optimal decision. However as noted in the press, many such borrowing countries have experienced debt crises, where some countries have been unable to make payments, and where the cost of servicing their debt can represent a significant portion of their GDP (and in some cases represents 20-40+% of government budgets) and hinder development.

a) Explain one factor that can contribute to debt crises for small open economies.

Factor:

b) Explain how recent increases in inflation, globally, have contributed to the debt crisis for these small open economies.

Explanation:

57. For each of the following ‘shocks’ to an economy, explain the impact on exchange rate for that country:

a) Large reduction in labour supply (relative to other countries) following the COVID pandemic.

Impact on e:

Explanation:

b) A reduction in Government Spending following the end of pandemic supports.

Impact on e:
Explanation:

58. Relative to the USD, the Canadian dollar has fluctuated but overall has been declining for about two decades. Some suggest that this decline is a good thing, and that the Bank of Canada should target a lower exchange rate in order to increase our exports and decrease our imports.

a) Explain how the Central Bank could aim to lower the exchange rate for the Canadian dollar. What policy could they use?

Policy:
Explanation:

b) Do our models suggest that exports increase and imports fall (NX rise) as a result of this policy?

Is NX predicted to rise:
Explanation:

c) What is the potential impact of this expansionary Monetary policy on foreign investment in Canada?

Predicted impact on Foreign investment:
Explanation:

59. Would the Bank of Canada be able to reduce the exchange rate and reduce inflation simultaneously using an expansionary monetary policy? Explain.

Answer:

Explanation:

60. Suppose that you have data on USD exchange rates, as well as data from the Economist's "Big Mac Index,"[‡] across several countries. Suppose that this report shows the price of a Big Mac in South Korea as 5200 Won. While the price of a Big Mac in the US is 5.58.

[‡] The Big Mac index shows the price of Big Macs in the local currency of several countries. The ratio of 'the price in local country in the local currency' to the 'price in USD in the US' generates the implied purchasing power of the USD in the local country.

a) What is the implied purchasing power of the USD in South Korea?

Answer:

Explanation:

b) If PPP holds, what would be implied the exchange rate be for South Korea?

Answer:

Explanation:

c) If the actual exchange rate is 1,274, does the Big Mac Index data suggest that the Won is overvalued or undervalued?

Answer:

Explanation:

d) Explain two drawbacks to using the Big Mac Index to estimate whether currencies are over/under valued.

Drawbacks:

e) Consider prices of diamonds, bananas, and cars across countries. Which price index measure might be more useful than the Big Mac to estimate over/under valuation of currencies, and why?

Answer:

61. Suppose that Mexican pesos depreciate in value relative to Canadian dollars. Specifically, suppose that in 2007 one dollar was worth 10 pesos, whereas in 2027 one CAD could purchase 15 pesos. Lets simplify things and assume that the only product produced in Canada and Mexico are churros.

a) If the price of a churro in Canada is \$1 in 2007, and 1.50 in 2027, what is inflation in Canada over this period? Show your calculation.

Answer:

Calculation:

b) If the price of a churro in Mexico was 5 pesos in 2007, and 15 pesos in 2027, what is inflation in Mexico over this period? Show your calculation.

Answer:

Calculation:

c) Does the depreciation of the Mexican Peso mean that it is cheaper for Canadians to visit (and eat) in Mexico in 2027 compared to 2007? Explain.

Answer:

Explanation:

62. Suppose the world is comprised of two large, open economy, countries: Florin and Guilder. And suppose that country Florin currently had a negative trade balance (imports>exports).

a) Is Florin a net lender or a net borrower in the world?

Answer:

Explanation:

b) If future marginal product of capital rises in Guilder, explain what will happen to the trade balance in each country.

Answer:

Explanation:

62. Suppose that there are only two countries in the world: Nelarn is a small open economy, and Orr is a large open economy. If Orr reduces taxes on investment income and Nelarn does not, what will be the impact on the net exports in Nelarn? (For this question, abstract from any secondary effects e.g. due to potential changes in exchange rates)

Answer:
Explanation:

63. The film “Commanding Heights” claims that Ronald Reagan's tax cuts got the U.S. economy out of stagflation (stagnant economy & inflation at same time).

a) Explain how a reduction in income tax, t (where $N^s = (1-t)w$) could reduce inflation and real interest rates, and increase employment and production in a market clearing long run framework. Use the Labour Market, Asset Market and Goods markets diagrams to help explain how the “Commanding Heights” assessment could be correct.

Labour Market	
Goods Market	Asset Market
Explanation:	

b) Assume that U.S. net exports fall as a result of Reagan's policy (Y effect dominates). Explain the impact of Reagan's policy on Canada's NX, GDP and e. (Assume US only trades with Canada)

Impact on the following variables (underline one):	Explanation:
NX (increase, decrease, same, ambiguous)	
GDP (increase, decrease, same, ambiguous)	
e (increase, decrease, same, ambiguous)	

c) Will the trade balance and the exchange rate always move in the opposite direction? Explain why or why not.

Answer:

Explanation:

64. Suppose the large open economy of Muggleland uses a contractionary monetary policy resulting in lower Y in Muggleland. Assume the Y effect dominates changes in NX in Muggleland and abroad.

a) How will Muggleland's monetary policy impact the economy of their trading partner Diagon Alley (a smaller country)? Specifically, how will Diagon Alley's Y , r , NX and e change? Use the a diagram for Diagon Alley's exchange market to help you explain what happens to e_D (where e_D represents the exchange rate for Diagon Alley's currency relative to the currency of Muggleland). Let D^S and D^D represent demand and supply for Diagon Alley's currency(dollars).

Diagon Alley's Exchange Market

Answer:

Explanation:

b) Now, instead of allowing e to fluctuate, assume that these countries were in a fixed exchange rate regime and Diagon Alley pegged their exchange rate to Muggleland. (Assume they want to keep it pegged at an official rate equal to e_{D1}^*). Would the impact of Muggleland's policy on Diagon Alley's GDP (Y) be larger or smaller? Explain.

Answer:

Explanation:

c) Would your answer from part b depend on how Diagon Alley fixed its exchange rate? Explain.

Answer:

Explanation: